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Enlightening Wisdom

Acropolis Institute of Technology & Research, Indore

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DRISHTI

Submitted to:

Department of Computer Science and Engineering

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Supervised by: Prof. Akshay Dubey

Team Members

1. Riya Kaushal
2. Reeti Shrimal
3. Priyal Waghmare
4. Mandisha Mirza
5. Priyanka Chirote

Project Presentation Outline

- Introduction
- Problem Statement
- Survey of Existing Systems
- Project Objectives
- Requirement Analysis
- System Designs/UML Diagrams
- Solution Proposed
- The Outcome Discussion
- Conclusions and Limitations

Introduction

Project Title:
Drishti – An AI-based Smart Surveillance System for Real-time Headcount and Recognition.
About the Project:
The system leverages Artificial Intelligence and Computer Vision to detect, count, and recognize individuals in real time. It uses **YOLOv8** for person detection and **DeepFace** for facial recognition, while a web-based dashboard displays live video feed, headcount, and alerts. Unknown individuals are labeled as “Unknown” to ensure privacy.
Solution Provided:
Drishti combines **computer vision, deep learning models, and a secure database** to deliver accurate and privacy-aware surveillance. It automates crowd monitoring, reduces manual supervision, and provides real-time insights through an interactive dashboard. Its modular architecture allows future scalability for multi-camera, cloud, and mobile-based integration.
Motivation:
Driven by the **limitations of traditional CCTV** and rising security needs, the project aims to provide **efficient, accurate, and scalable smart surveillance** for campuses, offices, and public spaces.

The Problem Statement

Traditional CCTV-based surveillance systems are **limited to passive video recording** and depend heavily on **manual monitoring by security personnel**. This leads to several issues such as:

1. Inefficient detection of unusual activity or overcrowding.
2. High chances of **human error and delayed response**.
3. Lack of **real-time alerts and intelligent analysis**.
4. Inability to **recognize individuals automatically** while ensuring privacy.
5. Limited scalability for modern **smart city and institutional needs**.

Thus, there is a strong need for an **AI-powered intelligent surveillance system** that can **automate real-time person detection, headcount, and facial recognition** while maintaining **privacy, accuracy, and scalability**.

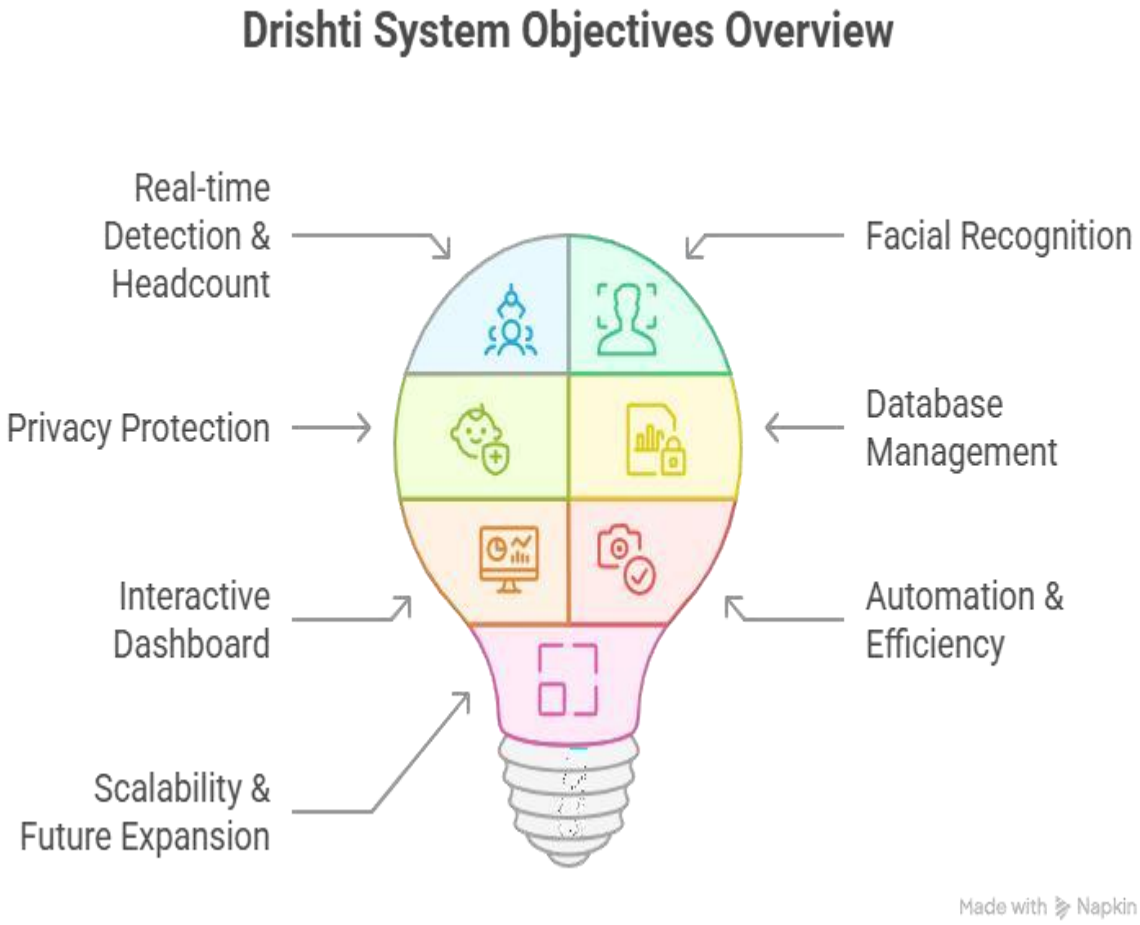
Survey of Existing Systems

S. No.	Project Title	Key Features / Description	Techniques Used	Domain / Application
1	Real-Time Human Detection and Tracking using YOLO and DeepSORT	Detects and tracks individuals in CCTV streams	YOLO, DeepSORT	Crowd monitoring, surveillance
2	COVID-19 Social Distancing Monitoring System	Counts people, measures distance for social distancing	YOLOv3, DeepSORT	Pandemic control, public safety
3	Smart Surveillance System using Face Recognition	Recognizes faces from pre-stored dataset	Haar Cascade, LBPH, DeepFace	Campus security, identity verification
4	Automated Attendance System using Face Recognition	Captures faces and marks attendance automatically	CNN, DeepFace	Schools, workplaces
5	Crowd Density Estimation using Computer Vision	Counts individuals to estimate density	Object detection, CNN	Event safety, public gatherings

Objectives

The primary objectives of **Drishti: An AI-based Smart Surveillance System** are:

- **Real-time Detection & Headcount** – To detect and count individuals using **YOLOv8** with high accuracy.
- **Facial Recognition** – To integrate **DeepFace** for recognizing known individuals and displaying their details.
- **Privacy Protection** – To label unrecognized individuals as **"Unknown"** without storing their data, ensuring data privacy.
- **Database Management** – To design a secure and scalable database for storing recognized identities and historical records.
- **Interactive Dashboard** – To develop a **web-based interface** that shows live video feeds, headcount, alerts, and recognition results.
- **Automation & Efficiency** – To reduce dependence on manual surveillance and minimize human error.
- **Scalability & Future Expansion** – To build a modular system that can later support **multi-camera setups, cloud storage, and mobile notifications**.



Requirement Analysis

1. Functional Requirements

- **Real-Time Person Detection:** Detect individuals in video frames using YOLOv8 and provide live headcount.
- **Facial Recognition:** Identify known individuals via DeepFace; unknown individuals labeled as “Unknown.”
- **Dashboard & Visualization:**
    1. Live video feed display.
    2. Real-time headcount and recognized individual details.
    3. Historical logs and statistics.
- **Database Management:** Secure storage of recognized individuals’ data, headcount logs, and timestamps.
- **Alerts & Notifications (Future Scope):**
- Overcrowding alerts.
- Mobile app notifications for security personnel.
- **System Scalability (Future Scope):** Multi-camera support, cloud integration, and large-scale deployment.

Continue

2. Non-Functional Requirements

- **Performance:** Real-time processing with minimal latency; optimized CPU/GPU usage.
- **Security & Privacy:**
  1. Unknown individuals' data is not stored.
  2. Secure communication via HTTPS/SSL.
  3. Optional dashboard access authentication.
- **Usability & Accessibility:** Intuitive and responsive dashboard; high-contrast colors, clear icons, and easy navigation.
- **Reliability & Availability:** Continuous operation without crashes; secure and consistent database storage.
- **Scalability & Extensibility:** Modular design to support future multi-camera setups, cloud storage, and mobile integration..

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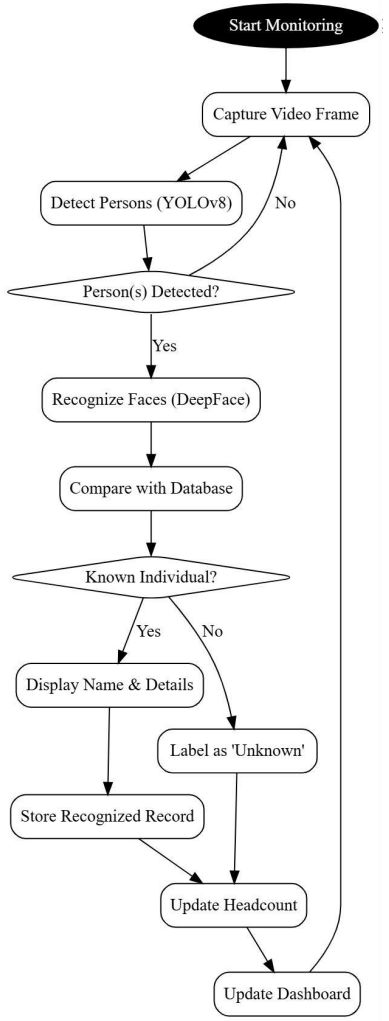
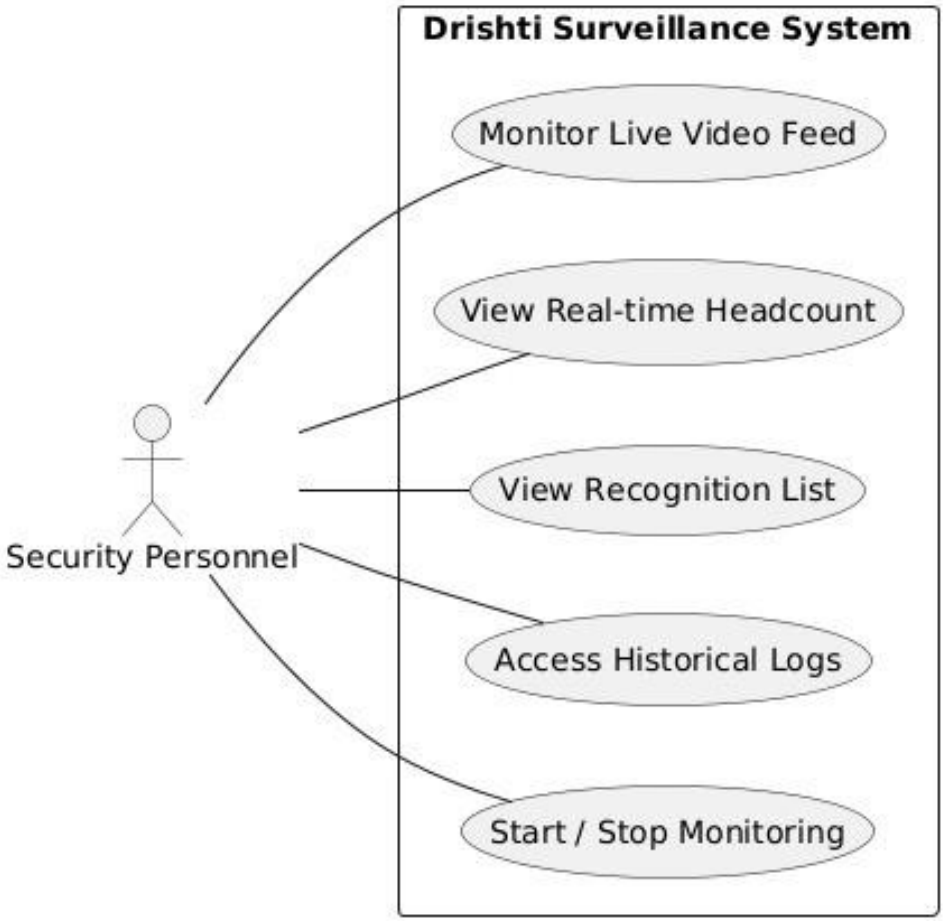
System Designs and UML Diagrams

- Flow Chart



Real-time Monitoring and Recognition

➢ Use case



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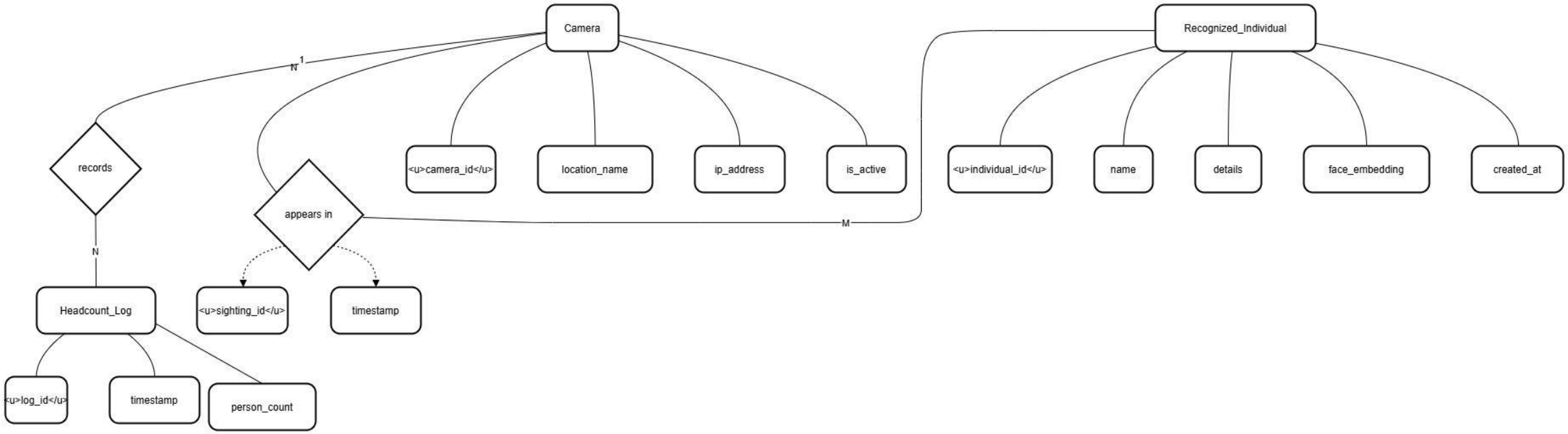
➢ ER Diagram



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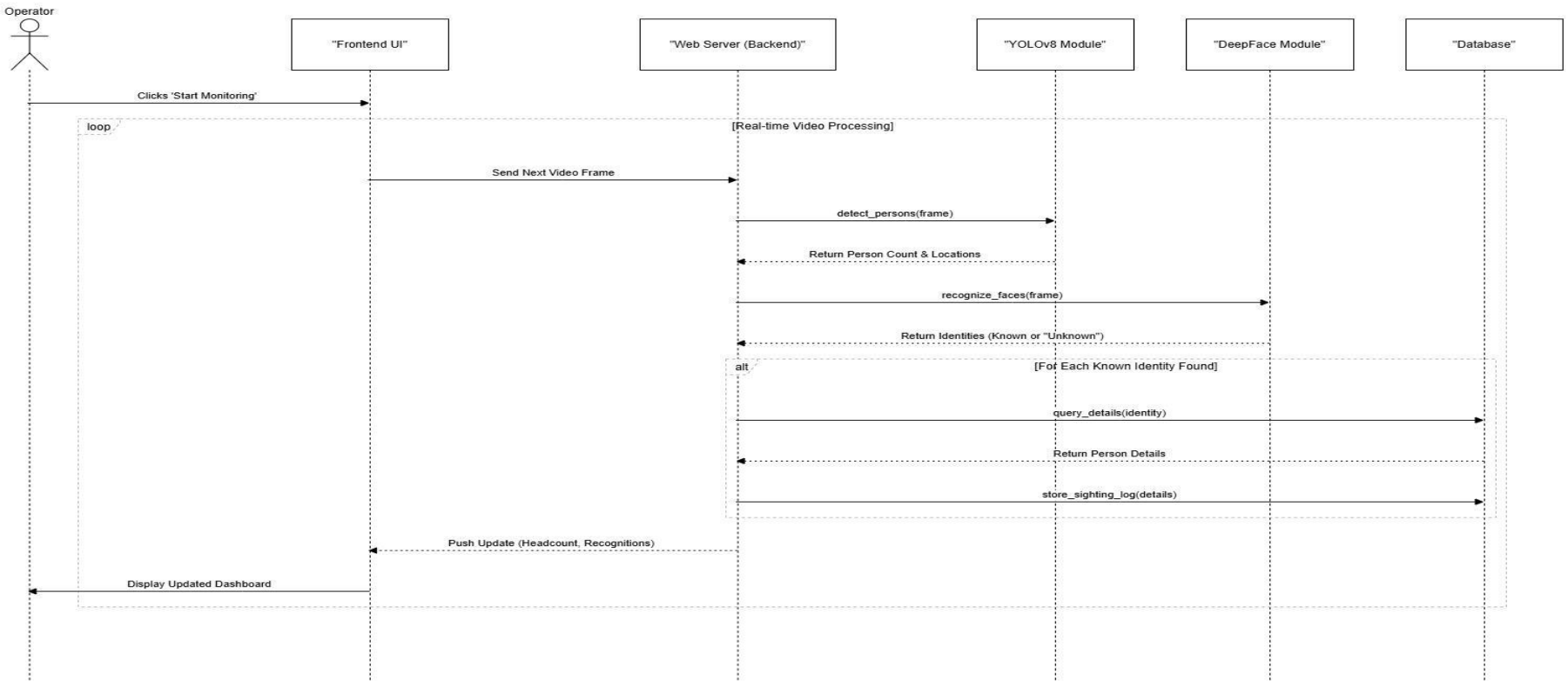
➢ UML Diagrams



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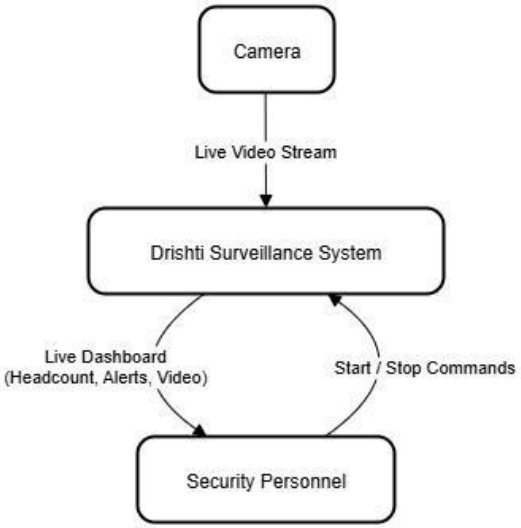
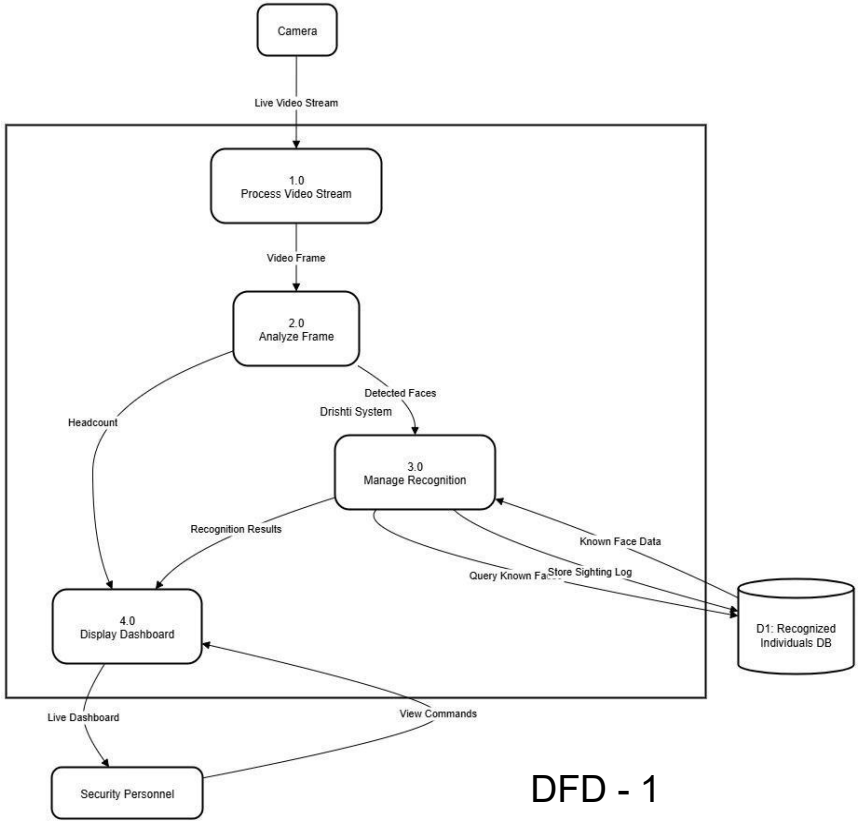
Data Flow Diagrams



DFD -0



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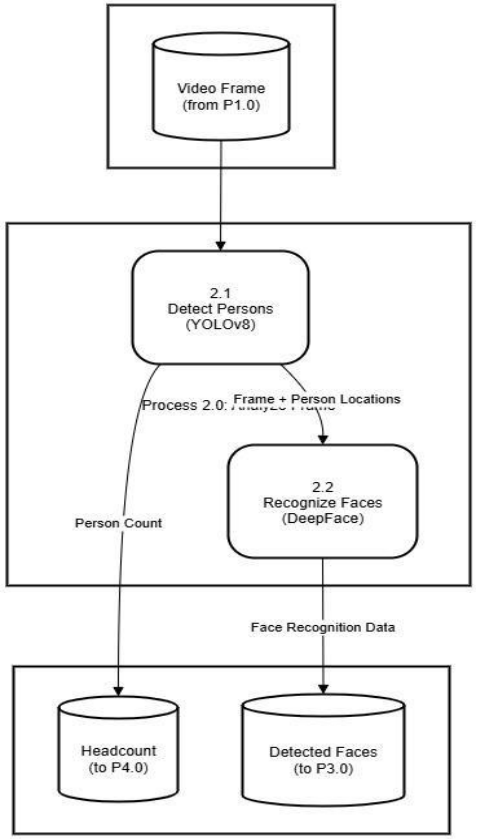

DFD - 2

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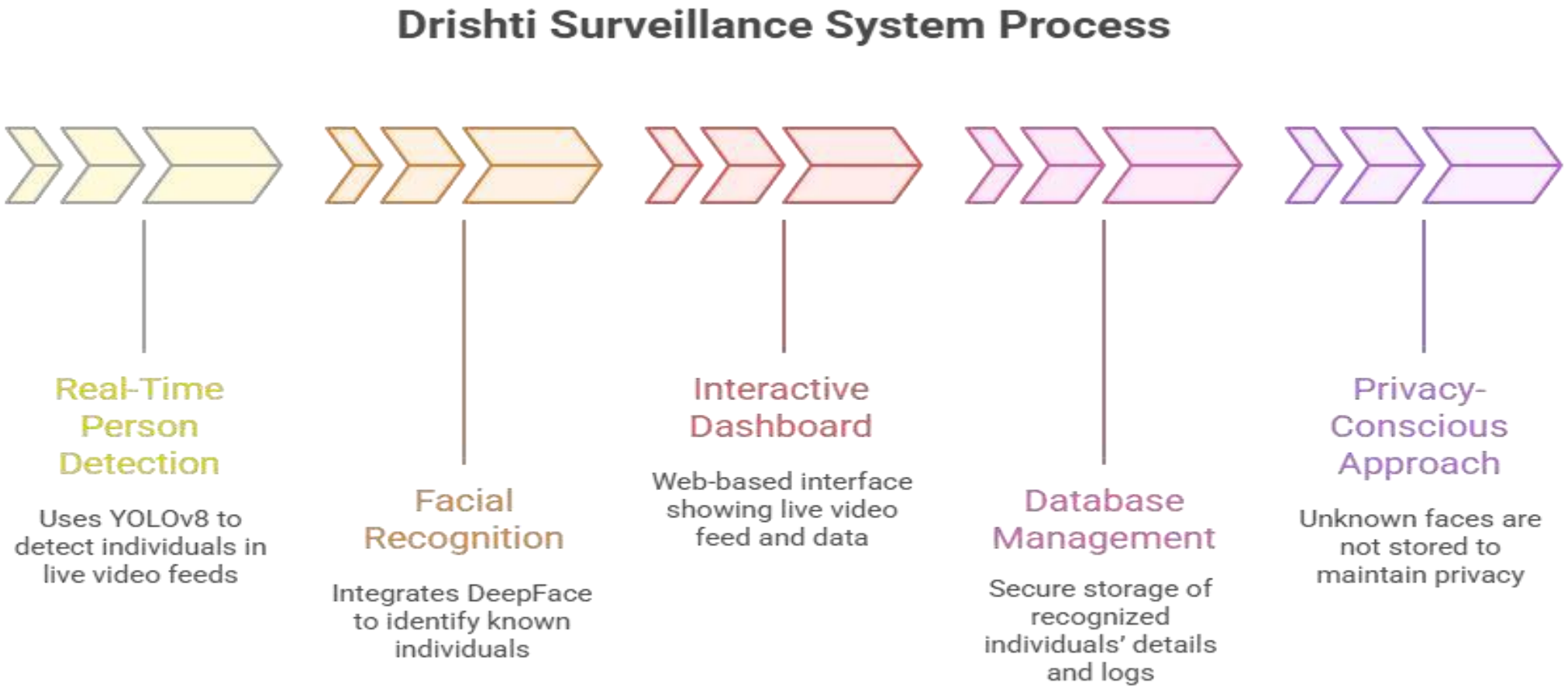
Solution Proposed

The proposed solution for Drishti is:



The Outcome Discussion

Expected Outcome of the Project Development :

- **Real-Time Person Detection:** Accurately detects and counts individuals in live video feeds.
- **Facial Recognition:** Recognizes known individuals while labeling unknown persons as "Unknown" to ensure privacy..
- **Interactive Dashboard:** Displays live video, headcount, recognized faces, alerts, and historical logs for efficient monitoring.
- **Automated Monitoring:** Reduces the need for manual supervision and enhances operational efficiency.
- **Data-Driven Insights:** Historical logs and headcount statistics support better decision-making and crowd management.
- **Enhanced Safety & Security:** Provides a reliable, real-time surveillance system suitable for offices, campuses, labs, and small public spaces.

Conclusion and Limitation

Drishti is an AI-based smart surveillance system that effectively integrates real-time person detection using YOLOv8 and facial recognition using DeepFace to provide efficient and intelligent monitoring. The system automatically counts individuals, manages crowd density, and identifies known persons while labeling unknown individuals as “Unknown,” ensuring privacy protection. Its interactive, web-based dashboard allows security personnel to monitor live video feeds, receive alerts, and access historical logs in real time, improving decision-making and operational efficiency. Designed with a modular and scalable architecture, Drishti is capable of future enhancements such as multi-camera support, cloud integration, and mobile notifications. Overall, it offers a reliable, privacy-conscious, and automated surveillance solution suitable for offices, campuses, labs, and small public spaces, laying the groundwork for advanced smart city security applications.

Limitations

- Environmental & Performance Factors: The system's accuracy is affected by external conditions such as lighting, camera quality, camera angle, resolution, and crowd density.
- Computational Cost: The system requires high computational resources (a powerful CPU or GPU) to perform real-time AI processing and analysis.
- Scope of Deployment: It is currently designed for small to medium-scale areas (like offices, classrooms, or labs) and is not suitable for very large spaces or large-scale deployment without optimization.
- Camera Support: The system is limited to working with only a single camera; it does not currently support multiple cameras.
- Recognition Dataset: Facial recognition is limited to the pre-built dataset of known individuals. The accuracy of this recognition depends heavily on the quality and size of that specific dataset.
- Data Storage: Unknown faces are not stored or added to the database to preserve privacy, meaning the system cannot "learn" new faces on its own.

Q&A

THANKS